Project Report: Intelligent Edge Computing Gateway for Industrial IoT

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1. Problem Definition:
Cloud-only IoT architectures suffer from high bandwidth costs and 250ms roundtrip latency.
This project develops an edge gateway for real-time anomaly detection in vibration sensors.

2. Technical Methodology:
We deployed quantized MobileNetV2 models on a Raspberry Pi 4 running FreeRTOS and Linux.
Sensors stream acceleration data at 1 kHz over an isolated RS-485 Modbus industrial bus.

3. Implementation and Evidence:
Custom PCB shield designed with galvanic isolation to protect against electrical surges.
Firmware processes 1,000 samples/sec with 14ms local inference latency.

4. Innovation:
Novel dynamic weight pruning algorithm reduces memory footprint by 42% with no accuracy loss.

5. Testing and Validation:
Validated across 500 operating hours on a test motor rig; achieved 97.8% F1-score for bearing fault d

6. Documentation and Reproducibility:
Complete KiCad PCB design files, Docker container, and C++ source code provided on GitHub.